

Explore Azure Databricks — Cheat Sheet

Microsoft Learn module: Explore Azure Databricks (Unit 1)

1. What Is Azure Databricks?

A cloud-based data platform, built on Apache Spark, that unifies data engineering, data science, and machine learning in a single workspace. It connects to sources across clouds (ADLS, S3, GCS) and enterprise systems (SAP, Oracle) so teams can integrate, transform, govern, analyze, and act on data.

- Core value: integrate data from many sources, engineer/transform raw data, store & govern it securely, apply real-time analytics/ML/AI, and drive better business decisions.

Data Lakehouse

- Blends data lake (scalable, cheap storage) with data warehouse (structure, performance) into one system — a single source of truth, avoiding duplicated/disconnected pipelines.
- **Medallion architecture**: a layered pattern where data is progressively cleaned and enriched as it moves through stages (raw → refined → curated).
- Two technologies power the Databricks lakehouse: **Delta Lake** (ACID transactions + schema enforcement) and **Unity Catalog** (unified governance for data & AI).

2. Get Started: Creating a Workspace

A workspace is your Azure Databricks deployment — a unified environment for a set of users. You can create one via:

- The Azure portal UI
- ARM, Bicep, or Terraform templates
- PowerShell: `New-AzDatabricksWorkspace`
- Azure CLI: `az databricks workspace create`

What you configure

- **Name and region** (check regional availability).
- **Pricing tier**: Premium (RBAC, Unity Catalog, SQL, Mosaic AI, serverless, Genie Code) or Trial (14-day free Premium trial).
- **Workspace type**: Serverless (pre-configured, managed storage — recommended for most cases) or Hybrid/Classic (compute & storage provisioned in your subscription — best for custom networking or on-prem connectivity).
- Optional **Managed Resource Group name** — auto-created RG holding the infra Databricks provisions behind the scenes.

Pricing tier update: the Standard tier stopped accepting new workspaces on April 1, 2026, and is being fully retired — existing Standard workspaces auto-upgrade to Premium by October 1, 2026. Premium is effectively the only tier going forward. A **Free Edition** (no time limit, daily usage caps, no classic compute) is also available for students and educators.

3. Navigating the Workspace UI

- **Sidebar** categories: Workspace (folders/notebooks), Recents, Catalog (Unity Catalog assets), Jobs & Pipelines, Compute, Marketplace.

- **By product area:** SQL (SQL Editor, Queries, Dashboards, Genie, Alerts, Query History, SQL Warehouses) · Data Engineering (Job Runs, Data Ingestion) · Machine Learning (Playground, Experiments, Features, Models, Serving).
- **+ New** creates workspace objects (notebooks, queries, repos, dashboards, alerts, jobs, pipelines, experiments, models, serving endpoints) or compute (clusters, SQL warehouses, ML endpoints).
- Top bar **search** finds notebooks, queries, dashboards, files, folders, libraries, UC tables, jobs, and repos in one place.

Genie Code (formerly Databricks Assistant)

- AI pair programmer built into notebooks, dashboards, and files — generates, explains, and fixes code/queries; diagnoses job issues; builds visualizations from natural-language prompts.
- **Agent mode** (GA for data science, data engineering, and dashboard authoring) lets it autonomously plan and complete multi-step tasks.
- Personalizes responses using Unity Catalog metadata (your tables, columns, descriptions).

4. Azure Databricks Workloads

Workload	What it's for
Data Engineering	Apache Spark big-data processing in the lakehouse; Python/R/Scala/SQL; pipeline building
Machine Learning	Build/train/deploy ML at scale; MLflow for the ML lifecycle; supports TensorFlow, PyTorch, Scikit-learn
SQL / BI	SQL warehouses, SQL editor, dashboards, auto-visualization for ad-hoc queries and reports (Premium tier required)
Streaming	Real-time / continuous data processing workloads

5. Key Concepts

Workspaces & Notebooks

- **Workspace:** secure, collaborative environment for notebooks, clusters, jobs, libraries, dashboards, experiments — accessible via web UI or REST APIs. Tied to Unity Catalog for governance; linked to an underlying (managed) resource group.
- **Notebooks:** interactive documents mixing runnable code + Markdown, multi-language via magic commands, real-time multi-user collaboration. Contain **code cells** and **Markdown cells**; can be version-controlled, scheduled as jobs, or exported.

Clusters & the two-layer architecture

- **Control plane:** Microsoft-managed backend services for your account.
- **Compute plane:** the external layer that actually processes data, living in your Azure subscription.
- A cluster = one **driver node** (coordinates execution) + one or more **worker nodes** (distributed computation). Can be fixed-size or auto-scaling.

Compute options

- **Serverless compute:** fully managed, on-demand, auto-scales — fastest startup, least management overhead.
- **Classic compute:** user-provisioned clusters — full control over VM sizes, libraries, runtime versions.
- **SQL warehouses:** compute optimized for SQL/BI, available as serverless or classic.

Databricks Runtime lifecycle

Customized Apache Spark builds with extra libraries/perf improvements. Each version follows a lifecycle:

- **Legacy** — available but not recommended
- **Deprecated** — marked for future removal
- **End of Support (EoS)** — no more patches/fixes
- **End of Life (EoL)** — retired, no longer available

Lakeflow Jobs

- Workflow automation/orchestration — a **job** is a container for one or more **tasks** (run a notebook, a Spark job, external code, etc.).
- Triggers: on a **schedule**, in response to an **event**, or **manually**.

Delta Lake

- Open-source storage framework adding transactional reliability on top of cloud object storage.
- Supports **ACID transactions**, scalable metadata (billions of files), **time travel**/versioning & rollback, and unified batch + streaming on the same table.
- **Delta tables are the default table format** in Azure Databricks.

Databricks SQL & SQL Warehouses

- Databricks SQL brings warehousing to the lakehouse: ANSI SQL editor, dashboards, BI tool integration. **Premium tier only**.
- All Databricks SQL queries run on **SQL warehouses** — scalable compute decoupled from storage.

Type	Photon	Predictive IO	Intelligent Workload Mgmt	Startup
Serverless	✓	✓	✓	~2–6 seconds
Pro	✓	✓	—	~4 minutes
Classic	✓	—	—	~4 minutes

- **Serverless:** compute lives in the Databricks account; best startup/scaling; ETL, BI, exploratory analysis.
- **Pro:** compute lives in your Azure subscription; use when serverless isn't available or you need custom networking (federation, hybrid/on-prem).
- **Classic:** compute in your subscription; entry-level performance; basic interactive exploration only.

MLflow

- Open-source platform managing the end-to-end ML lifecycle: experiment tracking, model management, reproducibility, deployment.
- Also supports generative AI workflows and evaluating/improving AI agents.

6. Data Governance: Unity Catalog & Microsoft Purview

Fragmented data (across databases, lakes, formats, and asset types — tables, files, models, notebooks, dashboards) creates compliance risk, vendor lock-in, and slower innovation. Unity Catalog + Microsoft Purview address this together.

Unity Catalog

- Centralized **access control, discovery, lineage, audit logs, and quality monitoring** across all workspaces in a region.
- **Metastore** = top-level metadata container, typically one per region, shared across multiple workspaces.
- **Three-level hierarchy:** catalog.schema.object — catalogs (teams/environments) → schemas (projects/use cases) → objects (tables, views, volumes, functions, models).

- **Managed tables:** UC handles governance *and* storage (always Delta Lake). **External tables:** UC governs access only; storage/lifecycle managed externally; supports Delta, CSV, JSON, Parquet, etc.
- Fine-grained access control via ANSI SQL (e.g. `GRANT CREATE TABLE ON SCHEMA mycatalog.myschema TO `finance-team``) from metastore level down to rows/columns.
- Discover data via Catalog Explorer — tags, comments, AI-generated descriptions, lineage, table insights, ER diagrams.
- Enabled by default on new workspaces in most accounts.

Microsoft Purview

- Cross-platform governance service spanning on-premises systems, multiple clouds, and SaaS — data discovery, classification, lineage, access governance.
- Integrated with Unity Catalog, Purview discovers Lakehouse data and ingests its metadata into the **Data Map**, giving one consistent governance layer across your whole environment.
- Can scan across public/private networks (via managed integration runtime), the entire UC metastore or selected catalogs, and even the legacy workspace-level Hive metastore.
- Auto-classifies sensitive data (built-in or custom rules), shows detailed lineage, and can run scans on-demand or on a recurring schedule.

7. Key Terms at a Glance

Term	Definition
Lakehouse	Combines data lake storage + data warehouse structure/performance into one system
Medallion architecture	Layered pattern that progressively refines data (raw → refined → curated)
Genie Code	AI pair programmer built into Databricks (formerly Databricks Assistant); has an autonomous Agent mode
Cluster	Driver node + worker node(s) providing the compute for a workload
Lakeflow Jobs	Orchestration: a job is a container for one or more scheduled/event/manual tasks
Delta Lake	ACID-transactional storage layer; default table format in Azure Databricks
SQL Warehouse	Scalable, storage-decoupled compute for Databricks SQL (serverless, pro, or classic)
MLflow	Open-source platform for the end-to-end ML (and GenAI) lifecycle
Metastore	Top-level Unity Catalog metadata container, usually one per region
Microsoft Purview	Cross-platform governance service that can ingest Unity Catalog metadata into its Data Map

Sources: Microsoft Learn training module "Explore Azure Databricks," cross-checked against current docs.microsoft.com/azure/databricks pages (SQL warehouse types, subscription/tier management) — verified July 2026.